7 LEARNING FROM THE CLIMATE OF THE DISTANT PAST

HOW MIGHT WE SEEK to understand climate? One way, of course, is to observe the system in its current state: we may determine radiative balance, investigate the way forcing factors and feedbacks operate, study the various components of the carbon cycle, and so on. From these and other activities, we gain an understanding of how the present climate system works. But this approach offers only a snapshot of a system that operates on a continuum of timescales extending far beyond the decades to centuries for which we have quantitative observations of climate.

To understand the present-day climate, we must also look to other times. What were the causes of the large climate swings that occurred many thousands and even millions of years ago? Are those causes relevant to climate change today? How were regional swings connected to events on a global scale, and how do past dynamics help us to understand present-day dynamics? Can we find events in the distant past that mimic our present predicament and thus provide insight into what the future may hold? Through investigations of such questions, it has become apparent that the study of past climate, or *paleoclimatology*, is a fertile field for our growing knowledge of present climate.

The Ice Age and Paleoclimatology

Earth's history has been characterized by long periods of stable climate punctuated by periods of fluctuating climate that last for millions to tens of millions of years. One such period appears to be the present *ice age*, which began about 2.6 million years ago.[1] The label "ice age" is somewhat misleading because it does not mean that the planet is either completely or perennially frozen. Rather, during this time, climate has fluctuated rapidly (in geological terms) between glacial intervals that continued for tens of thousands of years and shorter interglacial intervals. During the glacial times, average mid- and high-latitude land-surface temperatures were 5 to 15°C (9–27°F) lower than at present; temperatures during the interglacial times were not too different from those of today. The most recent glaciation reached its height about 21,000 years ago and then dissipated over thousands of years to bring us to the present interglacial, which began about 11,700 years ago. This interglacial period is known to geologists as the

Holocene epoch. The interglacial intervals have amounted to only about 10 percent of the past million years, indicating that today's interglacial conditions have been relatively short-lived.

The world during the glacial periods was a much different place than it is today. When the most recent glaciation was at its peak, sea level was 120 to 130 meters (400–425 feet) lower than it is now because so much water, instead of being in the ocean, was tied up as ice on the continents.[2] Indeed, ice blanketed 30 percent of the land area. It covered Scandinavia, the North Sea, most of the British Isles, and parts of northern Europe. In North America, ice covered everything north of an arc extended from Long Island in the east to central Illinois, across the Great Plains and the northern Rocky Mountains, and on to Puget Sound in the west. These ice sheets shaped much of the landscape we see today in these northern regions.[3]

THE BEGINNINGS OF PALEOCLIMATOLOGY

It can be reasonably argued that paleoclimatology as a systematic discipline had its origins in the recognition of Earth's most recent glacial period. By the end of the eighteenth and the turn of the nineteenth century, a number of naturalists had begun to recognize puzzling features suggesting that glaciers had been far more extensive in the past. Particularly in the

FIGURE 7.1 ***Pierre des Marmettes***
Pierre des Marmettes is a gigantic glacial erratic composed of Mont Blanc Granite. It was transported to its present location near Monthey, Switzerland, on the slope of the Valais (the valley of the Rhône upstream from Lac Lemans, where Geneva is located) by a glacier that flowed down the Val Ferret, a side valley. In this photograph, taken in 1929, the boulder is seen in the middle of a vineyard. Pierre des Marmettes figured prominently in early thinking about the origin and significance of glacial erratics. (L. Wehrli, image Dia_247-F-00312, archive of the library of the ETH Zürich, http://www.e-pics.ethz.ch/index/ethbib.bildarchiv/ETHBIB.Bildarchiv_Dia_247-F-00312_77864.html)

valleys of the Swiss Alps, naturalists and locals alike were struck by the presence of large boulders in odd places, such as in the flat, low expanses of river valleys or perched high on the side of the valleys (figure 7.1). Not only were their locations strange, but keen observers realized that the boulders were made not of rock found anywhere locally but generally of rock found far up the valleys. They began to wonder how the boulders could have been transported to their present locations. Deeply engrained was the belief that the biblical flood had given shape to much of the landscape. To most people, it was by this means that the boulders had arrived at their current positions. Recognizing the importance of glaciers as agents of transport, Charles Lyell (1797–1875), author of the text *Principles of Geology*, which profoundly influenced nineteenth-century geologic thought, sought to reconcile this realization with belief. He theorized that the boulders had been trapped in ice and that ice-bearing boulders had been washed down the valleys to their present sites by the flood. But there was no obvious evidence of a flood, especially one that had risen high up the sides of the valleys. To some, in fact, it seemed possible that the boulders had been carried by glaciers, remnants of which were (and still are) present at the head of some valleys. The locals certainly were familiar with the glaciers. They observed their waxing and waning; they saw rocks and other debris embedded in the ice; and they knew about the large, chaotic piles of gravel ridges (now known as moraines) associated with the glaciers (figure 7.2). They also saw rock surfaces scarred by strange, parallel grooves as if scraped by some enormous weight. The grooves are

FIGURE 7.2 Terminal moraine
Terminal moraines, such as this one in western Greenland, form at the noses of glaciers where the ice melts and deposits its entire heterogeneous, mixed load of boulders, gravel, silt, and other debris that it happens to be carrying. (E. Mathez)

termed striations, which we now know formed when ice-embedded rocks scraped across the surface of bedrock (figure 7.3). A few naturalists were even led to postulate that extensive parts of Europe had been covered by a vast ice sheet.

One of those naturalists was Louis Agassiz (1807–1873) (figure 7.4). Although not one of the idea's originators, he developed the hypothesis that Europe had once been covered by ice. He supported it with extensive observational evidence and thus was primarily responsible for its popularization and ultimate acceptance.[4] As the story goes, in 1837 Agassiz was a young professor at Neuchâtel and well known for his studies of fossil fishes. At the annual meeting of the Swiss Society of Natural Sciences, of which he had just been elected president, Agassiz shocked his colleagues by presenting a paper not about fossil fishes, as everyone expected, but about observations suggesting that ice had once covered the nearby Jura Mountains. He then put forth the radical notion that the Jura ice was but a remnant of an enormous polar ice sheet that had covered nearly all of Europe and much of North America. The notion received a frosty reception, and his colleagues remained skeptical, indeed hostile, even after Agassiz, the following day, took them into the nearby mountains to see the evidence with their own eyes. Undaunted, Agassiz published his ideas in *Études sur les glaciers* (*Studies on Glaciers*, 1840), a book that would become a classic and help to give birth to an entirely new field of research.

FIGURE 7.3 Glacial striations
The striations were carved as ice-carried boulders scraped across Manhattan Schist, the bedrock of Manhattan in New York City. The outcrop is known as Umpire Rock and is near the southwestern corner of Central Park. (G. Harlow, American Museum of Natural History)

FIGURE 7.4 Louis Agassiz
Louis Agassiz (1807–1873) was a pioneer in the study of ice ages. (Courtesy of South Caroliniana Library, University of South Carolina, Columbia)

In North America, geologists were to recognize and map four distinct layers of till (the general term used for the heterogeneous mixture of rock fragments of different sizes and shapes transported and deposited by glaciers) separated by fossil-bearing lake, fluvial, and peat deposits indicative of warmer climates, marking four major glacial advances. In many parts of Scandinavia, odd lines of gravel were recognized as raised beaches, relicts from a time when the land had been depressed by the great weight of overlying ice and was later exposed when the ice melted and allowed the land to rebound. In fact, parts of Scandinavia are still rising (chapter 10).

Earth's Orbital Characteristics and Milankovitch Theory

Observations indicating that ice age climate has fluctuated between cold glacial and warm interglacial periods (also termed stadials and interstadials) raise an obvious question: Why? What could possibly have caused the dramatic, back-and-forth swings of climate on such apparently short geologic timescales? The answer is *orbital forcing.*

Orbital forcing of climate is the principal mechanism that underlies what is known as *Milankovitch theory.* The notion that Earth's orientation relative to the Sun can affect climate was first advanced in 1842 by the French mathematician Joseph Adhémar (1797–1862). Nearly 80 years later, Serbian civil engineer and mathematician Milutin Milankovitch (1879–1958) quantified and formalized Earth's total orbital characteristics as a theory to explain glaciations (Historical Note).

HISTORICAL NOTE
Milutin Milankovitch, Celestial Mechanics, and Planetary Climates

Milutin Milankovitch (also Milankovic; 1879–1958) was born into a "venerable" Serb family in Dalj, a small village on the banks (now on the Croatian side) of the Danube River northwest of Belgrade. In secondary school, young Milutin distinguished himself in mathematics and, in 1896, entered the High Technical School in Vienna, where he majored in civil engineering. Seven years later, he defended his doctoral dissertation, the subject of which was the distribution of pressure in weight-bearing objects (for example, bridge abutments).

By 1909, at age 30, Milankovitch had been awarded six patents, was the highest-ranking engineer at a prestigious engineering firm in Vienna, and was widely recognized as an expert on reinforced concrete.[1] Then he quit. He quit to join the University of Belgrade as a professor of applied mathematics in a far less sophisticated city than Vienna and at a much lower salary. What's more, within just a few years, he wrote several completely original papers that would become the basis for an "astronomical theory of ice ages," a theory that eventually made his the most celebrated name in paleoclimatology.

Milutin Milankovich as a student in Vienna, ca. 1900. (Photographer unknown [public domain], Wikimedia Commons)

Why Milankovitch abandoned his wildly successful career as a civil engineer is not entirely clear. Whatever the reason, "he decided to commit himself to . . . climatology while reading [a paper on the] allocation of the Sun's heat on the surface of the Earth, in which the starting point was a wrong equation."[2] Fortunately, he chose to investigate one of the few subjects that was actually amenable to mathematical analysis. During his first three years at the University of Belgrade, he wrote several papers calculating the solar insolation reaching Earth and other planets and the temperatures of planetary surfaces. He also began thinking about temporal insolation changes as a possible answer to the still widely debated question of the cause of the ice age.

In an exquisite feat of timing, Milankovitch married Tinka-Hristina Topuzović on June 14, 1914, six weeks before the outbreak of World War I. The timing was auspicious for Milankovitch. He, like all Serbian citizens living in Austro-Hungary, was imprisoned by the military, but his new wife pleaded with one of his former colleagues and friends, Emanuel Czuber, to intercede on his behalf. Through Czuber's efforts, Milankovitch was allowed to pass his captivity in Budapest. He was also permitted to work, helped by the fact that the head of the library of the Hungarian Academy of Sciences granted him complete access to the library. Milankovitch remained in Budapest throughout the war, returning to Belgrade and university life in 1919.

In Budapest, Milankovitch wrote several papers and completed his first monograph, *Théorie mathématique des phénomènes thermiques produits par la radiation solaire* (*Mathematical Theory of Thermal Phenomena Caused by Solar Radiation*, 1920). In it, he introduced the "curve of insolation," which shows how solar insolation reaching Earth has changed over the past 130,000 years in response to secular changes in Earth's orbital elements. The work soon attracted the attention of climatologists, motivating Milankovitch to extend the curve back to 650,000 years. The work also showed how insolation varies with latitude. Over the next two decades, Milankovitch developed his ideas further—for example, by examining how changes in insolation affect planetary ice cover.

Realizing that the various ideas and calculations on which his astronomical theory of climate change was based were scattered through many publications written over several decades, Milankovitch decided in 1939 to collect all the relevant material into a single, comprehensive publication. Thus began his supreme work, *Kanon der Erdbestrahlung und seine Anwendung auf das Eiszeitenproblem* (*Canon of Insolation of the Earth and Its Application to the Problem of the Ice Ages*). (Here *canon* means "general laws.") The 626-page book "is a comprehensive mathematical picture of the effect of the Sun's radiation on planets' solar climates."[3] It almost did not survive: the factory where the book had been printed was destroyed in the bombing of Belgrade in April 1941, but nearly all the already printed pages had been moved to a mostly undamaged print warehouse. The work was translated into English in 1969.

In 1941, war intruded on Milankovitch's life for the second time. He withdrew from university activities during the German occupation of Yugoslavia, but was reinstated as a professor after the war. Milankovitch continued to advocate his theory until his death in Belgrade in 1958. The theory was never fully accepted in his lifetime, a common criticism being that the orbitally induced variations in insolation were too small to account for the glacial–interglacial cycles. That changed in 1976 with publication of the classic paper by James Hays, John Imbrie, and Sir Nicholas Shackleton that showed that deep-sea sediment cores display evidence for variations in global ice volume consistent with Milankovitch cycles.[4] Today, Milankovitch theory is almost universally accepted.

Milankovitch was posthumously honored twice—in 1970 and 1973—by the International Astronomical Union, which named a crater on the far side of the Moon and another on Mars after him.

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1. This historical note is based primarily on the "virtual exhibition" "Milutin Milankovic: Famous Scientist from Belgrade University," http://arhiva.unilib.rs/unilib/eng/about_us/exhibitions/milankovic_virtual/index.php; A. Petrović and S. B. Marković, "Annus Mirabilis and the End of the Geocentric Causality: Why Celebrate the 130th Anniversary of Milutin Milanković?," *Quaternary International* 214 (2010): 114–118; the brief biography "Milutin Milankovic," European Geophysical Society, http://www.egu.eu/egs/milankovic.htm; and "Milutin Milanković," Wikipedia, https://en.wikipedia.org/wiki/Milutin_Milanković.

2. "Milutin Milankovic: Famous Scientist from Belgrade University."

3. Petrović and Marković, "Annus Mirabilis and the End of the Geocentric Causality."

4. J. D. Hays, J. Imbrie, and N. J. Shackleton, "Variations in the Earth's Orbit: Pacemaker of the Ice Ages," *Science* 194 (1976): 1121–1132.

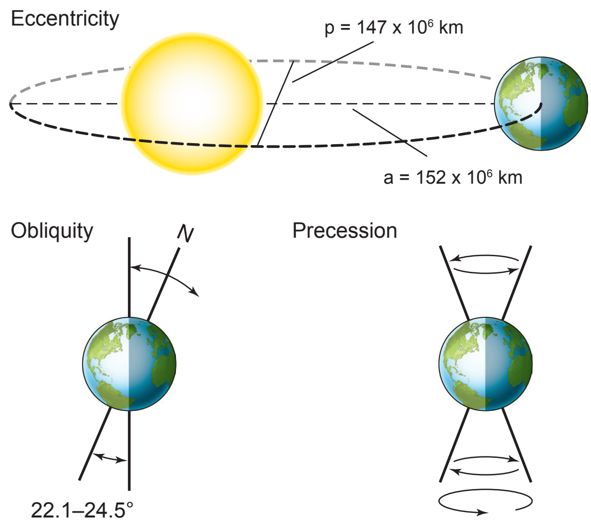

FIGURE 7.5 Earth's orbital changes: precession, obliquity, and eccentricity
The lengths of the present-day perihelion and aphelion, which define the eccentricity of Earth's orbit, are shown. For clarity, the Sun is displaced much farther from the center of mass of its eccentric orbit with Earth. (Redrawn from R. A. Rohde, Global Warming Art)

Three characteristics of Earth's orbit change in cycles of different lengths (figure 7.5). First, the tilt of the axis of rotation relative to the solar plane, or *obliquity*, which is currently 23.5 degrees, changes from 22.1 to 24.5 degrees and back again approximately every 41,000 years. Earth's obliquity results in the seasons. Summer occurs in the hemisphere that tilts toward the Sun relative to Earth's rotational axis, and winter occurs in the hemisphere that tilts away from the Sun. If the tilt, or obliquity, of Earth is toward its maximum, the high-latitude winters and summers of both hemispheres are more intense, as are the equator–pole temperature gradients. The Northern Hemisphere, however, is much more affected than the Southern Hemisphere because it has more extensive high-latitude landmasses that can accommodate ice sheets.

Second, Earth's orbit around the Sun becomes periodically more or less elliptical with a main component of 413,000 years and lesser components of 95,000 and 125,000 years, resulting in an apparent but approximate 100,000-year periodicity. The *eccentricity*, *e*, describes the departure of an orbit from a circular path and can be calculated as

$$e = (ra - rp) \,/\, (ra + rp),$$

where *ra* is the radius at the farthest distance from the center of mass of the ellipse,[5] and *rp* is the radius closest to the elliptical center of mass. At present, $ra = 152 \times 10^6$ kilometers, and $rp = 147 \times 10^6$ kilometers (see figure 7.5). This results in the current value of $e = 0.0167$ and an Earth–Sun distance variation of 3.34 percent during the course of a year. The corresponding annual variation in solar radiation reaching Earth is 6.68 percent. The farthest distance of Earth from the Sun in its orbit is known as the *aphelion*; the shortest is termed the *perihelion*. While these variations in eccentricity are important for climatic fluctuations over hundreds of thousands of years, they are not responsible for seasonal changes, a commonly held misconception.

The eccentricity varies from about 0.0034, when the orbit is nearly circular, to 0.058, when it is slightly elliptical. The greater the eccentricity, the more the solar-radiation flux received by Earth varies over the course of the year. In fact, at maximum eccentricity, the radiation flux at the perihelion is 20 to 30 percent greater than that at the aphelion. Today, Earth's orbital eccentricity is low, so solar-radiation flux varies over the year by less than 7 percent.

The influence of eccentricity is partially mitigated by the fact that Earth's orbital velocity slows as it moves away from the Sun and increases as it moves closer. Thus the season that occurs when Earth is at its aphelion is longer than the season in the same hemisphere that occurs at the perihelion. For example, today the Northern Hemisphere summer occurs when Earth passes through the aphelion and winter occurs when it passes through the perihelion, and for this reason the Northern Hemisphere summer is longer by several days than its winter.

Third, Earth's axis of spin *precesses*, or itself slowly rotates around another axis that is observable by Earth's changing relation to the fixed stars. The periodicity of the precession is about 26,000 years.[6] When precession points the North Pole toward the Sun at the perihelion, the North Pole also points away from the Sun at the aphelion. The result is that the Northern Hemisphere experiences larger seasonal contrasts, and the Southern Hemisphere smaller seasonal contrasts, than in other orientations. The opposite is true when precession points the South Pole toward the Sun at the perihelion. Commonly, precession and eccentricity are combined into a *precession index* that describes how the seasons move into and out of phase with eccentricity.

The interplay of the three orbital cycles, according to Milankovitch theory, creates significant variations in insolation during the summer in the high latitudes of the Northern Hemisphere and accounts for the waxing and waning of the great ice sheets of the Pleistocene. The reason that summer rather than winter insolation is important in waxing and waning is that summer melting is far more efficient at removing ice than winter accumulation is at creating it. Milankovitch theory gained currency in the 1970s with the discovery that the chronology of climatic variations recorded in deep-sea sediment cores extending back almost 450,000 years could be accounted for by periodic forcings that occur at intervals of 23,000, 42,000, and 100,000 years.[7] Numerous other observations support the theory. For example, the 41,000-year oscillations in early Pleistocene climate are well explained by obliquity,[8] and evidence of Milankovitch cycles has been found in older sedimentary sequences as well. Perhaps the most definitive test of Milankovitch theory comes from the timing of the glacial cycles in the different hemispheres; the end of the most recent glacial cycle was in full swing 15,000 years ago, corresponding to an increase in northern latitude summer insolation. The warming in Antarctica lagged behind the change in insolation, however, implying that insolation change in the Northern Hemisphere is the fundamental driver of the ice age cycles.[9]

That said, Milankovitch theory is not without its problems. For example, it is also widely recognized that orbital parameters alone are not sufficient to explain the glacial and interglacial cycles because the orbital-induced variations in summer insolation at the northern latitudes are too small to account for the large swings of climate that characterize the cycles.

There must be some amplifier(s) (feedbacks [chapter 6]) of the orbital forcings internal to the climate system. Let us now look at the record in some detail to see what more it tells us.

Five Million Years of Climate

The record of climate during the Pliocene (5.33–2.59 million years ago) and Pleistocene (2.59 million–11,600 years ago) epochs is fairly complete.[10] This record has emerged primarily from the study of oxygen isotopes in deep-sea sediments. Because marine sediments are such important climate records and oxygen isotopes represent one of the principal tools for their study, the behavior of the isotopes is discussed in box 7.1.

MARINE SEDIMENTS AND THE OXYGEN-ISOTOPE RECORD

Every year, 6 to 11 billion metric tons of sediment accumulate on the ocean floor. The sediment consists mainly of the shells of dead planktonic (near-surface-dwelling) and benthic (deep-water-dwelling) organisms plus sand and mud from the continents. The marine-sediment record, which is determined by the systematic study of drill cores recovered from numerous parts of the ocean (figure 7.6), is rather complete, although burrowing organisms and ocean currents have in places disturbed the record.

One of the reasons that deep-sea sediments are so important is that they help to relate what was happening in the atmosphere to what was happening in the oceans. To cite some examples, the abundances and community makeup of certain microorganisms in marine sediments are proxies for sea-surface temperature (SST). Sediments in the North Atlantic Ocean also contain conspicuous layers of poorly sorted lithic fragments (heterogeneous rock fragments ranging from silt to boulders). This material was eroded and scraped off the continents by glaciers, carried to their present ocean locations in icebergs, and deposited as the icebergs melted. The layers define *Heinrich events*, times of abrupt collapse of Northern Hemisphere ice shelves and/or ice sheets, leading to rapid and voluminous discharges of icebergs into the North Atlantic.[11] Layers of eolian (wind-borne) sediment generally reflect dry conditions and strong winds, both indicative of cold climate. In fact, the evidence for Pleistocene cycling between glacial and interglacial intervals was first observed in deep-sea sediment cores as alternating carbonate-rich layers, representing the warm part of the cycle, and eolian layers of mud, representing the cold part.

Several key chemical characteristics of foraminifera and other microorganisms that accumulated over time on the ocean floor serve as proxies of past climate. One is the ratio of certain organic molecules in the lipid remains of planktonic foraminifera (those that live near the surface), resulting in a technique known as *alkenone paleothermometry*.[12] Alkenones are "biomarker molecules," meaning that they hold biological and biochemical information. Their relative abundances in the organic fraction of marine sediments have become important proxies for sea-surface temperature.

The remains of benthic foraminifera (that is, foraminifera that live in the deep ocean) tell a different story. Particularly useful have been the $\delta^{18}O$ (see box 7.1) of their carbonate shells, which depends on the $\delta^{18}O$ and temperature of the seawater in which they grew. Assuming

BOX 7.1
Oxygen Isotopes 101

Oxygen exists as three stable isotopes. Oxygen-16 (^{16}O) is the most abundant isotope (99.8 percent of all oxygen); the other two are oxygen-17 (^{17}O) and oxygen-18 (^{18}O) (table B7.1.1). The isotopes are said to *fractionate* from each other, which means that two phases—that is, physically distinct (solid, liquid, gas) forms of matter—acquire slightly different concentrations of the different oxygen isotopes as they equilibrate with each other.

Take water, for example. The vapor that forms by evaporation of the liquid possesses a slightly lower $^{18}O/^{16}O$ ratio than the liquid. This isotopic fractionation is a consequence of the differing nuclear masses of the isotopes, which influence characteristics that are dependent on mass, such as molecular binding energy. The magnitude of the fractionation depends on the ratio of the masses in question (that is, ^{18}O and ^{16}O, which differ from each other by two mass units, fractionate more from each other than do ^{18}O and ^{17}O, which differ by one mass unit).

Ratios of isotopes, such as $^{18}O/^{16}O$, are typically expressed in the so-called delta (δ) notation,

$$\delta^{18}\text{O (per mil)} = \frac{(^{18}\text{O}/^{16}\text{O})_{\text{sample}} - (^{18}\text{O}/^{16}\text{O})_{\text{standard}}}{(^{18}\text{O}/^{16}\text{O})_{\text{standard}}} \times 10^3$$

or the equivalent

$$\delta^{18}\text{O (per mil)} = \left[\frac{(^{18}\text{O}/^{16}\text{O})_{\text{sample}}}{(^{18}\text{O}/^{16}\text{O})_{\text{standard}}} - 1\right] \times 10^3$$

In this notation, the isotope ratio is not expressed as an absolute value but as a ratio relative to a standard. (A standard is a material of known composition, which for oxygen isotopes is commonly standard mean ocean water [SMOW] or, in analysis of marine carbonates, the carbonate fossil PeeDee Belemnite.)[1] Thus when the unknown and the standard have exactly the same $^{18}O/^{16}O$, then the unknown $\delta^{18}O = 0$, and if the $^{18}O/^{16}O$ of the unknown is either

TABLE B7.1.1 THE NATURAL ABUNDANCES (ATOMIC PERCENT) OF STABLE ISOTOPES MOST WIDELY USED IN PALEOCLIMATOLOGY

Hydrogen	$^{1}H = 99.985\%$	$^{2}H = 0.015\%$
Carbon	$^{12}C = 98.93\%$	$^{13}C = 1.07\%$
Nitrogen	$^{14}N = 99.63\%$	$^{15}N = 0.37\%$
Oxygen	$^{16}O = 99.76\%$	$^{18}O = 0.037\%$ (also ^{17}O)
Sulfur	$^{32}S = 95.00\%$	$^{34}S = 4.22\%$ (also ^{33}S, ^{36}S)

(Continued next page)

(Continued)

greater than or less than that of the standard, then its $\delta^{18}O$ is greater than 0 or less than 0, respectively. The notations for the isotopic ratios of the elements listed in table B7.1.1 are exactly analogous, with the only difference being that the standards are different. The factor 10^3 means that the $\delta^{18}O$ is expressed in parts per 1,000 (per mil), instead of, say, in the more familiar parts per 100 (percent).

For the same reason that water vapor has a slightly lower $\delta^{18}O$ than the water from which it evaporates, liquid droplets have a slightly higher $\delta^{18}O$ than the vapor from which they condense. To understand the relevance of this to climate studies, consider the evaporation of water from the equatorial ocean and what happens to it as it progresses poleward, cools, partially condenses, and then returns to the ocean from the atmosphere by precipitation. Figure

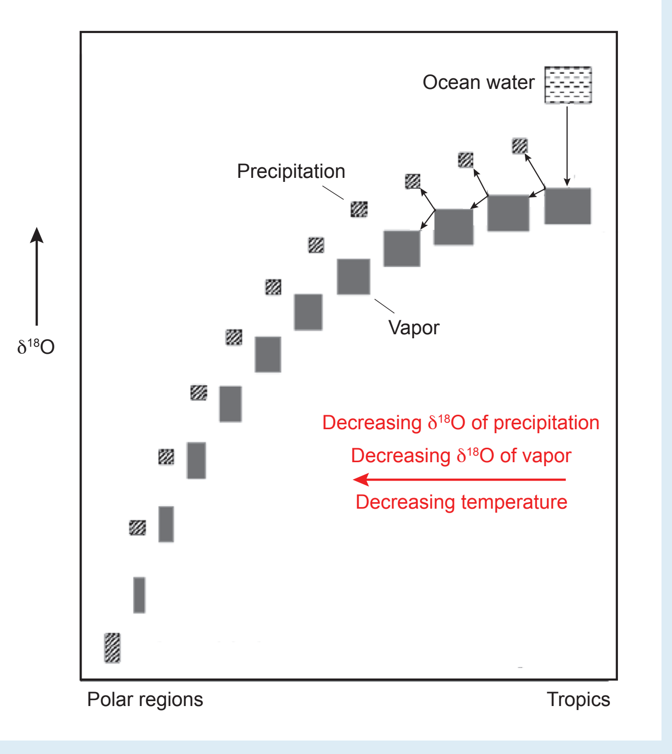

FIGURE B7.1.1

The progressive decrease in the $\delta^{18}O$ of water vapor and liquid water that condenses from the vapor and falls as precipitation with latitude. The change in $\delta^{18}O$ reflects the fractionation of the isotopes between liquid and vapor. Note that the fractionation of oxygen isotopes between water vapor and liquid water increases as temperature decreases. The oxygen-isotopic fractionation between water vapor and liquid water accounts for the lower $\delta^{18}O$ of the polar ice sheets compared with that of the ocean. (Redrawn from W. G. Mook, *Environmental Isotopes in the Hydrological Cycle: Principles and Applications*, vol. 1, *Introduction: Theory, Methods, Review* [Vienna: International Atomic Energy Agency, 2001], http://www-naweb.iaea.org/napc/ih/IHS_resources_publication_hydroCycle_en.html)

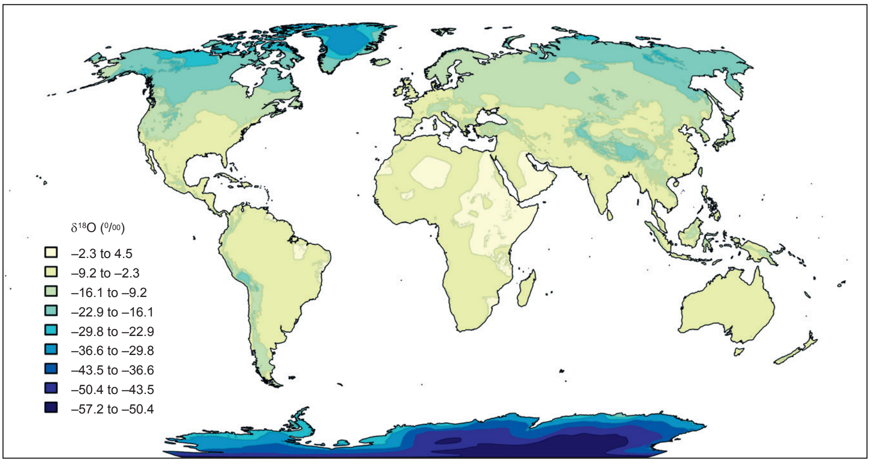

FIGURE B7.1.2
The global variation in the $\delta^{18}O$ of precipitation with latitude. Note the particularly low values in Antarctica and Greenland. (G. Bowen, University of Utah, WaterIsotopes.org, http://wateriso.utah.edu/waterisotopes/index.html)

B7.1.1 shows the initial formation of water vapor (*shaded box, far right*) near the equator with a lower $\delta^{18}O$ than the ocean (*lined box, far right*) from which it evaporated. Then, as temperature decreases, a small fraction of that mass of vapor condenses to form liquid water (*small box*) with higher $\delta^{18}O$. This leaves a second residual, now slightly smaller box of water vapor with a lower $\delta^{18}O$. As this process of distillation continues as the air mass moves poleward, one can see that both the residual boxes of water vapor and the liquid water separating from them display progressively lower $\delta^{18}O$ values. In the real world, this is manifest as a systematic change in the $\delta^{18}O$ of precipitation with latitude (figure B7.1.2).

Two important corollaries emerge from this. First, the $\delta^{18}O$ of Greenland and Antarctic snow (and thus ice) is considerably lower than the $\delta^{18}O$ of seawater. Therefore, during cold periods when a relatively high proportion of the total global surface water is held in the ice caps, the $\delta^{18}O$ of the ocean is greater than it is during warm periods, when less water is in the ice caps. In this way, the isotopic record of ocean water, such as that documented by certain fossils in marine-sediment stratigraphic sequences, constitutes a history of global temperature. Second, the atmosphere loses a higher proportion of its moisture by precipitation before reaching the high latitudes during cold periods than during warm periods. Therefore, the $\delta^{18}O$ of polar precipitation in times of cold is lower than in times of warmth. Such variations appear in stratigraphic sequences of ice.

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1. Expressing concentration relative to a standard derives from the fact that in mass spectrometry, the method by which isotope ratios are measured, the raw analytical signal from an unknown is an arbitrary number that has meaning only when expressed relative to the signal from a standard.

FIGURE 7.6 ***The drilling ship*** **JOIDES Resolution**
Many marine-sediment cores have been recovered as part of international ocean-drilling programs designed to explore the ocean floor using drilling ships. The tower midship is a drill rig that allows for drilling into the deep-ocean floor. Drill cores are recovered in 9.5-meter (31-foot) lengths, cut into 1.5-meter (4.9-foot) sections, and then cut in half lengthwise. Half the core is used for immediate investigation, while the other half is stored for future research. (W. Crawford, International Ocean Discovery Program, http://iodp.tamu.edu/publicinfo/drillship.html)

that the temperature of the deep ocean remains approximately constant at several degrees Celsius through swings of global climate (an assumption that is not always correct), variations in the $\delta^{18}O$ of benthic foraminifera reflect variations in seawater $\delta^{18}O$ rather than temperature. Seawater $\delta^{18}O$ depends on the proportion of global water held in the ocean relative to the ice caps. Thus the $\delta^{18}O$ of benthic foraminifera in a sequence of marine sediments provides a temporal record of global ice volume, a proxy for mean global surface air temperature. A complete benthic $\delta^{18}O$ record from 5.3 million years ago to the present is shown in figure 7.7. This record makes clear that conditions during the Pliocene were warmer than they are today, that since about 3 million years ago climate has been cooling, and that fluctuations in climate during the Pleistocene, especially in the last million years, have been much greater than during the Pliocene.

PLIOCENE WARMTH AND THE SHIFT TO PLEISTOCENE GLACIAL CYCLES

The so-called Pliocene warm period, loosely the time interval between 5 and 3 million years ago, has engendered considerable interest because it may represent an analogue for the present-day climate, given that its atmosphere contained about the same amount of carbon dioxide as today's. The global temperature during the Pliocene was 2 to 3°C (3.6–5.4°F) warmer and the sea level up to 12 to 32 meters (40–105 feet) higher than today.[13] Ice covered eastern Antarctica,[14] while ice caps were absent from the Northern Hemisphere.[15] Low-latitude SST

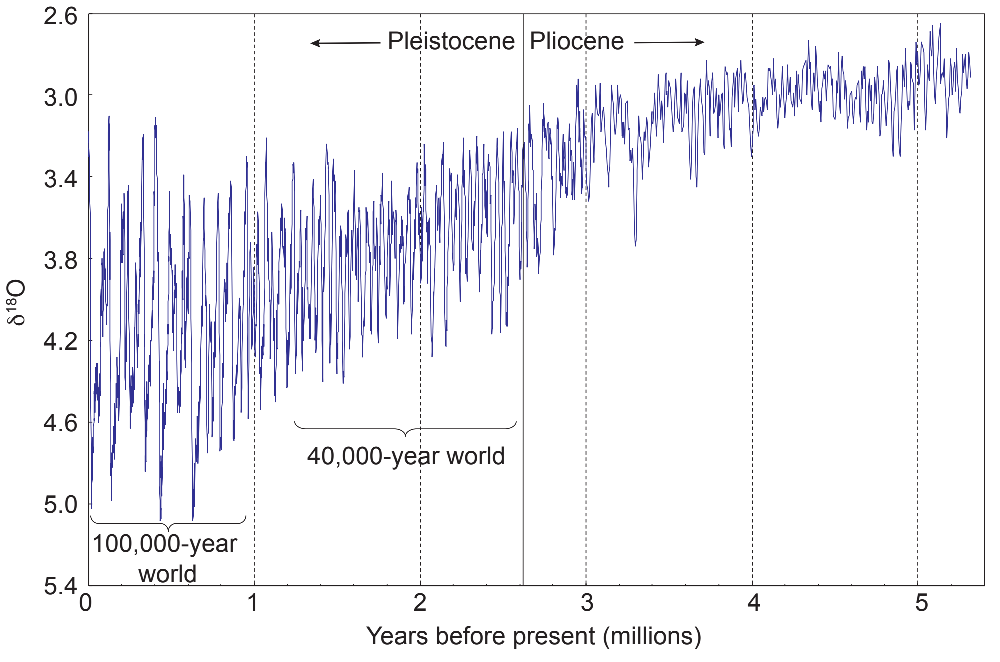

FIGURE 7.7 The oxygen-isotope record of seawater, 5.3 million years ago to the present

The oxygen-isotope record is based on the carbonate shells of benthic foraminifera. This is a "stacked" record in which data from different locations are correlated in time and then combined into a single record. (Data from A. V. Fedorov et al., "Patterns and Mechanisms of Early Pliocene Warmth," *Nature* 496 [2013]: 43–49, where references to the original sources may be found)

was only slightly warmer than it is today, but mid- and high-latitude SST was much warmer, resulting in lower meridional temperature gradients and less seasonality.[16] In the eastern equatorial Pacific, El Niño–like conditions were the norm, along with attendant El Niño–like weather patterns over land areas.[17] The tropical ocean thermocline was also much deeper than it is today, preventing tropical upwelling of deep, cold water.[18]

All this is curious: Why was early and mid-Pliocene climate so different from today's, while the external forcings were not so different? For example, solar-radiation flux was approximately similar, and a recent estimate puts the CO_2 content of the Pliocene atmosphere at about 400 parts per million (ppm), similar to today's value.[19]

Related to this question is a second perplexing one: Why did the climate change during the Pliocene? Beginning about 3.1 million years ago, the meridional temperature gradient began a gradual but substantial increase.[20] Conditions in the tropics did not change much, as evidenced by the fact that tropical SST remained approximately constant (and even today is little changed from its Pliocene temperature) (figure 7.8). Rather, mid- to high-latitude ocean-surface temperatures decreased 4 to 8°C (7–14°F), Arctic land temperatures plunged by as much as 19°C (34°F), a Northern Hemisphere ice cap developed for the first time in nearly 200 million years,[21] the Southern Hemisphere ice cap gradually expanded, and the ocean's equatorial warm pool contracted around the equator. Atmospheric circulation also changed, as the Hadley cells and subtropical highs intensified and migrated toward the equator. The increase in meridional SST gradient caused the tropical thermocline to shoal,

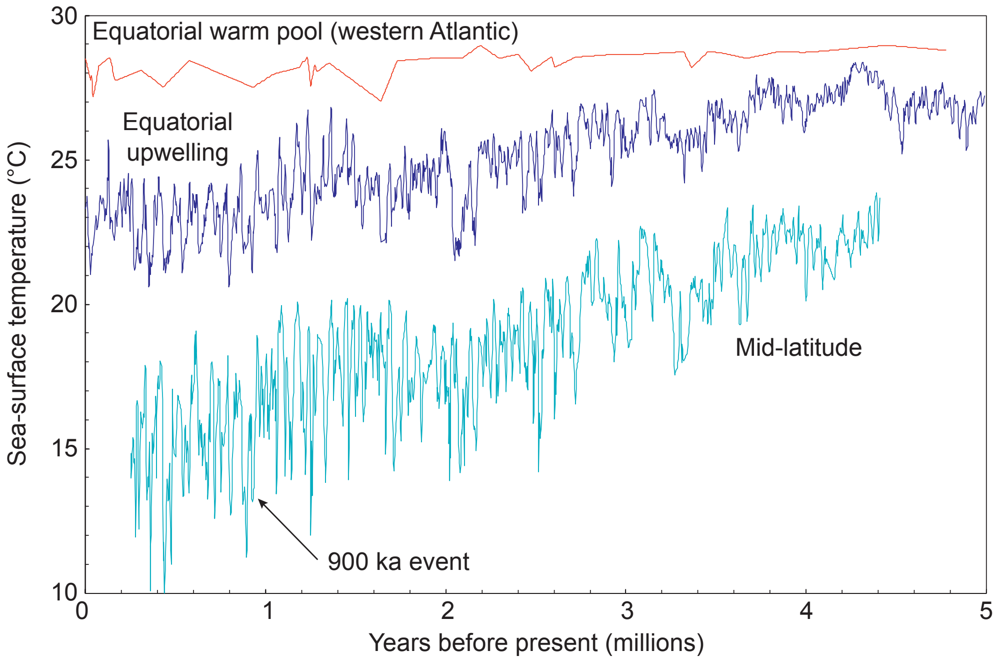

FIGURE 7.8 The Pliocene–Pleistocene record of sea-surface temperature

The sea-surface temperature record is from the western equatorial Atlantic Warm Pool (*red* [Ocean Drilling Project hole 925]), eastern equatorial Pacific region of upwelling (*blue* [ODP 846]), and mid-latitude Atlantic (*blue-green* [ODP 607]). The record illustrates that since the Pliocene, conditions in the tropics have changed little, but the mid- and high latitudes have experienced a long-term cooling, resulting in an increased meridional temperature gradient and greater seasonal contrast in these regions. The more robust upwelling in certain equatorial regions indicates that Walker circulation (chapter 3) has become more intense. (Data from Fedorov et al., "Patterns and Mechanisms of Early Pliocene Warmth")

allowing for the more frequent upwelling of cold water. Consequently, surface ocean cold tongues established themselves in the eastern equatorial Pacific and Atlantic oceans, intensifying Walker circulation and creating El Niño–like variability. These changes proceeded gradually until, at the end of the Pliocene, the climate abruptly shifted into a state whereby the ice sheets began to advance and retreat with a periodicity of approximately 41,000 years, reflecting amplification of the obliquity cycle (see figure 7.7).

Differences in orbital parameters are simply insufficient to account for the difference between the climate in Pliocene and that in the Holocene. Various other mechanisms have been proposed, but the reasons remain unclear. The mechanism(s) that brought about the cooling and the emergence of the "40,000-year world" also remain uncertain.[22]

PLEISTOCENE CLIMATES

Gradual cooling, decline in atmospheric CO_2 content, and obliquity-dominated glacial–interglacial cycles continued through the Pleistocene until about 1.2 million years ago. At that time, a slow transition began from the 41,000-year to the approximately 100,000-year cycles. The latter would dominate the Pleistocene from about 800,000 years ago onward (see figure 7.7). The mid-Pleistocene Transition (MPT), as it is called, is well documented in a number of signals, one of the most complete being SST.[23] This record is characterized by a strengthening of the cooling beginning about 1.2 million years ago, followed by an even more pronounced cooling between about 940,000 and 870,000 years ago, an interval known as the 900 ka event (see figure 7.8). The decline in SST occurred almost everywhere except in the Western Warm Pool of the equatorial Pacific Ocean. Meanwhile, the $\delta^{18}O$ of glacial-period benthic organisms gradually increased through most of the MPT and then more rapidly during the 900 ka event (this is particularly apparent in the temperature-corrected $\delta^{18}O$ record), suggesting first slow and then more rapid growth of polar ice sheets (see figure 7.7).[24] The fundamental cause of the MPT remains unclear. Like the emergence of the 40,000-year world at the beginning of the Pleistocene, it cannot be attributed to orbital parameters, which show no significant changes through the transition or even through the 900 ka event.

The past 800,000 years have been characterized by glacial periods that last, on average, about 80,000 to 90,000 years and that alternate with interglacial periods of about 10,000 years (figure 7.9). The glacial periods ended relatively rapidly; the interglacial periods ended more gradually, mostly by an initially rapid cooling followed by a more gradual one. This cycle happened again and again. The interval has seen eight prolonged glaciations spaced about 100,000 years apart (plus two shorter glaciations, depending on how they are defined).

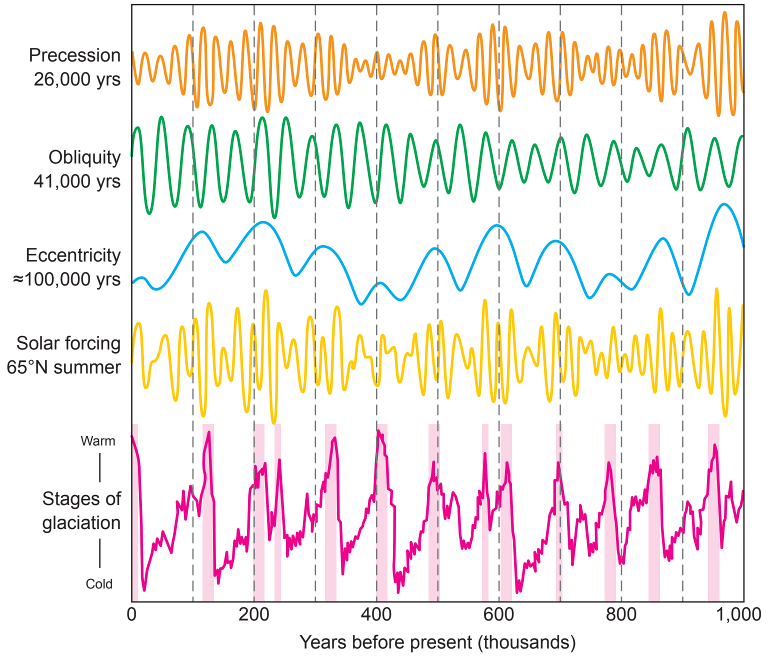

FIGURE 7.9 A million years of precession, obliquity, eccentricity, and solar insolation at 65°N latitude compared with the glacial stages

The glacial stages are based on the marine sediment $\delta^{18}O$ record. According to Milankovitch theory, variations in Earth's orbit were largely responsible for the cycles of climate change during the Pleistocene because of their effects on high-latitude summer insolation in the Northern Hemisphere. The cycles of precession and obliquity are more important than that of eccentricity. Alone, they are too small to account for the observed differences between glacial and interglacial periods, so other drivers in the climate system must have amplified the orbital-driven changes in insolation. (Redrawn from R. A. Rohde, Global Warming Art)

Why is the 100,000-year cycle so prominent in the ice age climate records, and how is it that the glacial terminations can be so abrupt if the orbital forcing is so small?[25] One idea is that the 100,000-year cycles are not related to eccentricity but to obliquity and precession. Indeed, a robust statistical argument has been made that the sudden glacial terminations of the past 700,000 years were actually associated with every second or third obliquity maximum magnified by precession, and that the eccentricity had little influence.[26] The deglaciations occur *on average* every 100,000 years, but their frequency actually varies such that they occur closer to alternating 80,000- and 120,000-year intervals than to 100,000-year intervals. Again, however, the good statistical correlations between orbital forcing and glacial cycles do not tell us why the shifts from glacial to interglacial periods have been so large and abrupt.

PLEISTOCENE ATMOSPHERE AND THE AMPLIFICATION OF ORBITAL FORCINGS

This brings us to the record of CO_2 in the atmosphere during the Pleistocene. Ice cores drilled into the East Antarctic Ice Sheet provide a continuous record of the greenhouse gas content of the atmosphere for the past 800,000 years (figure 7.10).[27] The record is derived from analyses of air trapped in bubbles in the ice (see chapter-opening figure). Ice forms as the snow laid down each year compacts and recrystallizes during burial. Newly fallen snow is porous and mixed with air. As the snow compacts and recrystallizes, some of the air is trapped to form bubbles.

Perhaps the most remarkable feature of the ice-core record is that for the past 800,000 years, the level of atmospheric CO_2 has varied in lockstep with the glacial–interglacial cycles. Furthermore, the CO_2 signal lags behind the temperature signal by centuries, although the lag is near the limit of the temporal resolution of the ice-core record and cannot be seen in figure 7.10. Nonetheless, the lag suggests that a buildup of greenhouse gases in the atmosphere did not initiate the warming that ended each glacial period. Rather, it appears that the warming was initiated by orbital forcings and then was *amplified* by a rise in the CO_2 content of the atmosphere. This would also explain the abruptness of the glacial–interglacial transitions. The exact mechanism linking atmospheric CO_2 content to ice age temperature variations is not known. Almost certainly, however, it has something to do with the exchange of CO_2 between the atmosphere and the Southern Ocean. This theory, as we will see, is supported by the more highly resolved climate record accompanying the termination of the last glacial period.

Another remarkable feature of the ice-core record is that the CO_2 content of the atmosphere never fell below 170 ppm during glacial periods or rose above 300 ppm during interglacial periods. This suggests that a specific set of processes has governed the amount of CO_2 in the atmosphere by transferring carbon back and forth between the atmosphere, on the one hand, and the ocean and land reservoirs, on the other. Finally, these and other data noted earlier show that the amount of CO_2 in the present-day atmosphere (~404 ppm at the beginning of 2017) is substantially higher than at any time during the past 800,000 years, and probably since at least the mid-Pliocene. An unanswered but important question is whether the natural processes that have kept the CO_2 content of the atmosphere in a narrow range

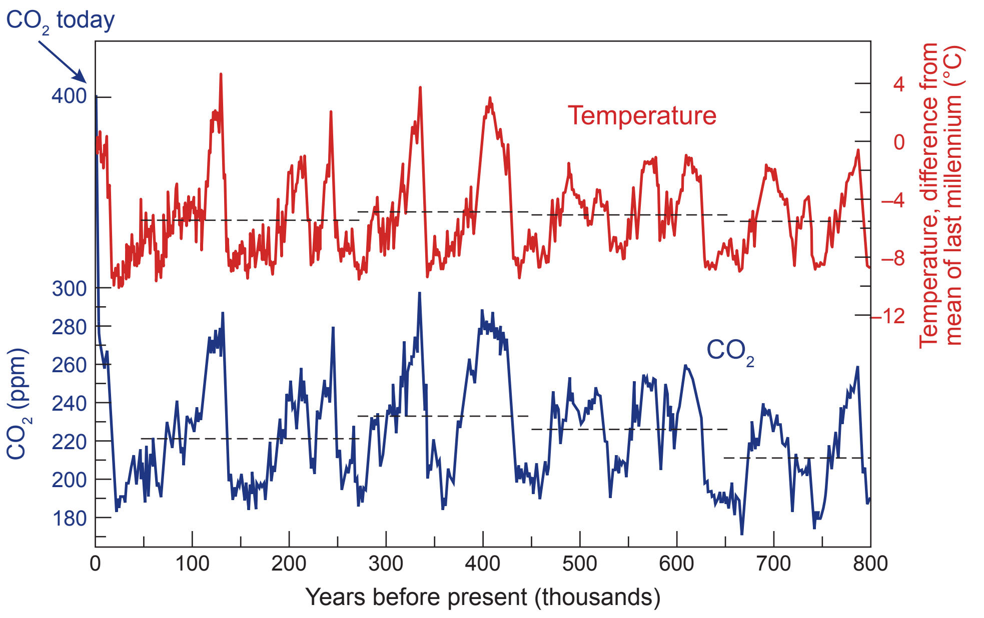

FIGURE 7.10 The 800,000-year record of atmospheric CO_2

The atmospheric CO_2 record is based on analyses of air trapped in Antarctic ice cores and compared with the temperature record derived from the ratio of hydrogen-2 to hydrogen-1 ($^2H/^1H$) in the ice. Throughout the entire 800,000 years, the CO_2 content of the atmosphere changed in concert with the glacial cycles and remained between 170 and 300 ppm. Since 2015, the average annual CO_2 content of the atmosphere has been above 400 ppm, higher than at any time in at least the past 800,000 years and probably since the Pliocene. (Redrawn from D. Lüthi et al., "High-Resolution Carbon Dioxide Concentration Record 650,000–800,000 Years Before Present," *Nature* 453 [2008]: 379–382)

for the past 800,000 years will continue to operate in the same way and with the same effect in today's high CO_2 world.

The record of methane (CH_4) is similar to that of CO_2. Over the past 800,000 years until the beginning of the industrial era, the CH_4 content of the atmosphere ranged from 0.320 to 0.790 ppm. Again, the present amount of CH_4 (1.83 ppm) is higher than at any other time in the past 800,000 years. Atmospheric CH_4 is believed to mirror the extent of natural wetlands, the main source of CH_4, and is high during warm periods and low during cold periods.

Finally, the ice-core samples have yielded a partial record of atmospheric nitrous oxide (N_2O). It also follows the trend of the other gases and is now higher than at any other time in the past 800,000 years.[28]

One Hundred Thousand Years of Climate Change

Among the most interesting records of climate are those from the Greenland Ice Sheet (figure 7.11). The ice represents an archive of unparalleled resolution through the last glacial period and the interglacial period preceding it. The latter is known as the *Eemian*, and it lasted from about 136,000 to 115,000 years ago. The record reveals an interglacial during which conditions were not too different from those of today and a glacial period of unexpected variability. In combination with Antarctic ice and marine-sediment cores from

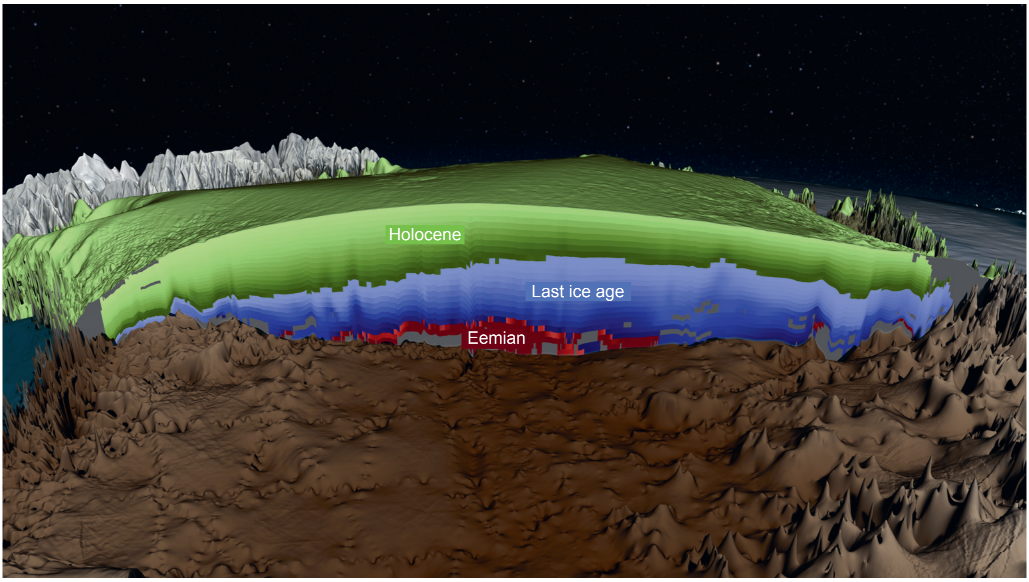

FIGURE 7.11 ***Cross section through the Greenland Ice Sheet***
The cross section shows layers of ice formed in the Holocene (the past 11,700 years), the last glacial and glacial termination (115,000–11,700 years ago), and the Eemian (136,000–115,000 years ago). (C. Starr, in "Greenland Ice Sheet Stratigraphy," January 23, 2015, NASA, Scientific Visualization Studio, https://svs.gsfc.nasa.gov/cgi-bin/details.cgi?aid=4249)

around the globe, it documents the termination of the last glaciation and provides insight into fundamental aspects of the climate system.

THE GREENLAND ICE CORES

In 1989, two teams, one European and the other American, began drilling into the Greenland Ice Sheet (figure 7.12).[29] By the summer of 1993, they had recovered two complete drill cores through the entire ice sheet. The American Greenland Ice Sheet Project Two (GISP2) drill hole penetrated 3,053 meters (10,016 feet) and the European Greenland Ice Core Project (GRIP) drill hole penetrated 3,029 meters (9,940 feet) of ice before hitting bedrock. It quickly became apparent that the recovered cores held a continuous record of climate that covered more than 100,000 years.[30] The record amazed many scientists because it revealed climate oscillations of far greater magnitude and rapidity than anyone imagined to have occurred during that period or that have occurred since the beginning of human agricultural activity. Later, North GRIP (NGRIP [1999–2003]) penetrated deep enough to span the beginning of the Eemian before reaching bedrock at a depth of 3,085 meters (10,122 feet),[31] and the North Greenland Eemian Ice Drilling project (NEEM [2008–2012]) reached past the Eemian into the previous glaciation before it encountered bedrock at 2,540 meters (8,333 feet) (figure 7.13).[32]

GISP2, GRIP, and NGRIP are on the top of the Greenland Ice Sheet at elevations of more than 2,900 meters (9,500 feet) above sea level. The average annual temperature on this lofty plateau is a frigid −31°C (−24°F), and the ice is well below the freezing point all the way down

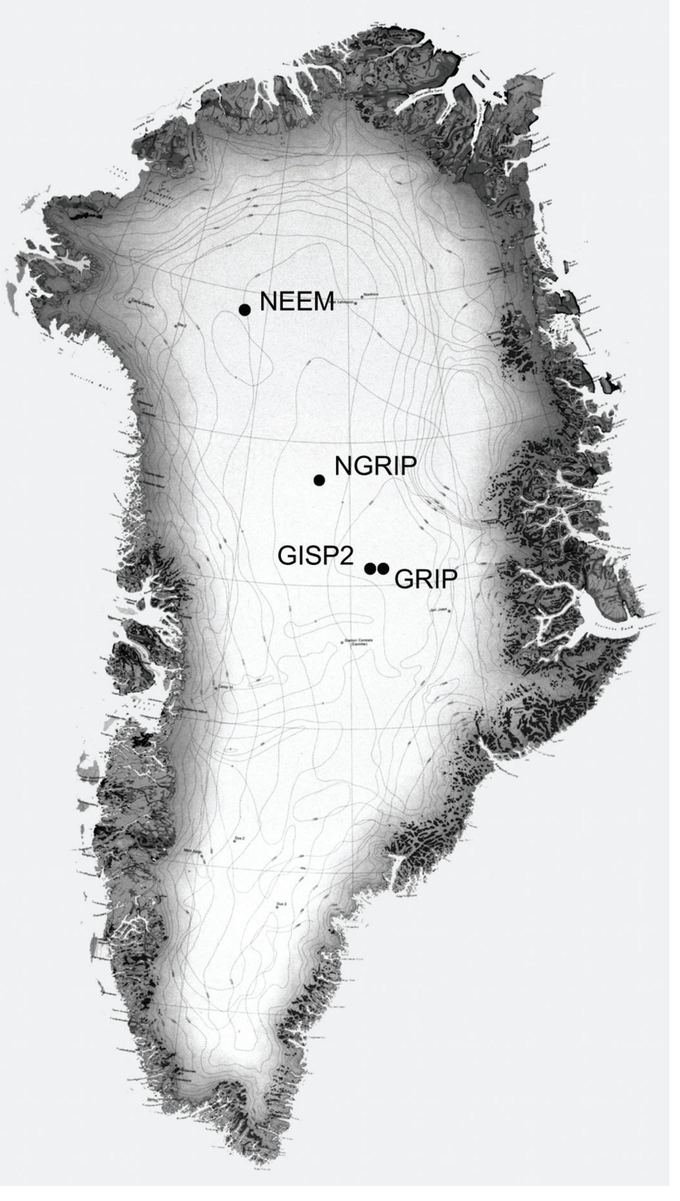

***FIGURE 7.12** Map of Greenland, showing the locations of ice cores that penetrated through the ice sheet*
Together, the cores from these locations provide a climate record that extends back more than 136,000 years through the last interglacial period, known as the Eemian. (Geological Survey of Greenland)

to bedrock, which eliminates any possibility that the climate record was disturbed by recent melting. The sites were chosen for an additional reason: they are very close to the *ice drainage divide* (the location where ice flows in opposite directions to the seas on either side of Greenland) and to where the bedrock is nearly flat. This choice was important. On a slope, ice slowly flows and may fold and fracture in the process, disrupting the delicate annual layers and destroying or repeating parts of the record. The proximity of the GISP2 and GRIP holes was

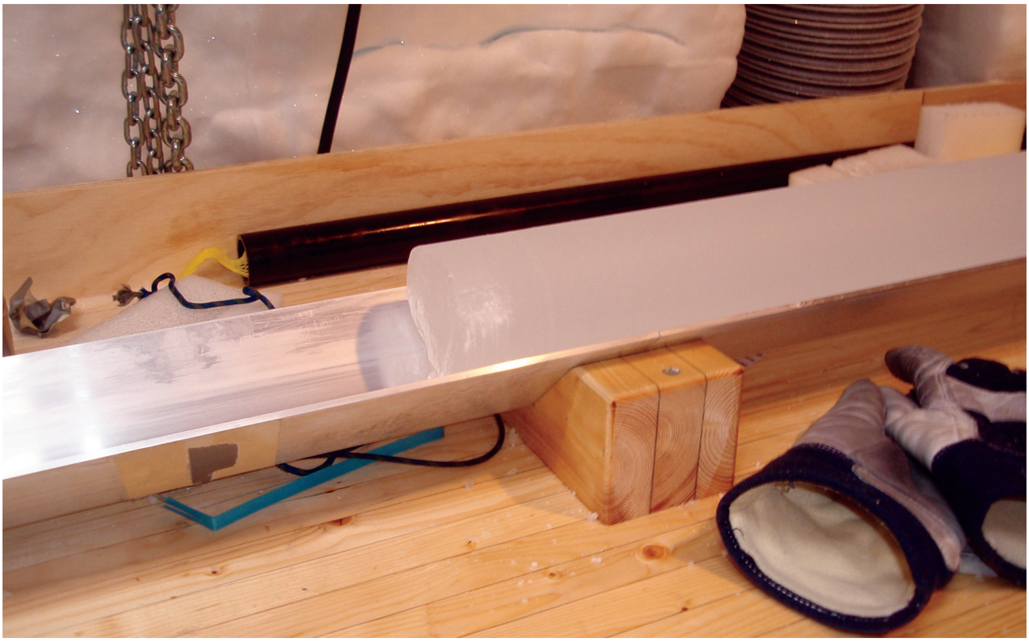

FIGURE 7.13 **Ice core**
This core was recovered from the NEEM drilling site in northwestern Greenland. (M. Murray, University of Calgary)

also purposeful: it allowed stratigraphic correlations to be made, thus helping to identify regions of the cores that were deformed. Fortunately, the disruptions were found to be limited to the cores' lowest 200 meters (700 feet), leaving undisturbed the upper 2,800 meters (9,200 feet). Above 2,800 meters, the GISP2 and GRIP records match each other almost perfectly.

The basal ages of the cores remain unknown, leading to the question of how long the present Greenland Ice Sheet has existed. Did it mostly disappear during the Pleistocene interglacials, or did remnants of it survive? The base of the GISP2 core for about 13 meters (42 feet) above its contact with the bedrock is contaminated with sediment, probably representing old soils entrained in the ice. This material has been found to contain high concentrations of certain *cosmogenic isotopes* (isotopes produced in the atmosphere and on Earth by the interactions of their molecules with cosmic rays), indicating that the sediment had been exposed to the atmosphere for many tens of thousands of years or more.[33] As we have seen, long interglacial periods are absent in the Pleistocene, and the last long interval of warmth was in the Pliocene more than 2.7 million years ago. Therefore, ice at the GISP2 location must have been present for at least that long.

CLIMATE SIGNALS IN THE ICE

The ice-core record is written in the language of geochemistry. The $\delta^{18}O$ of the precipitation that fell on Greenland during cold periods is lower (it has a lower $^{18}O/^{16}O$ ratio) than that of the precipitation that fell during warm periods, and in this way oxygen isotopes are a direct

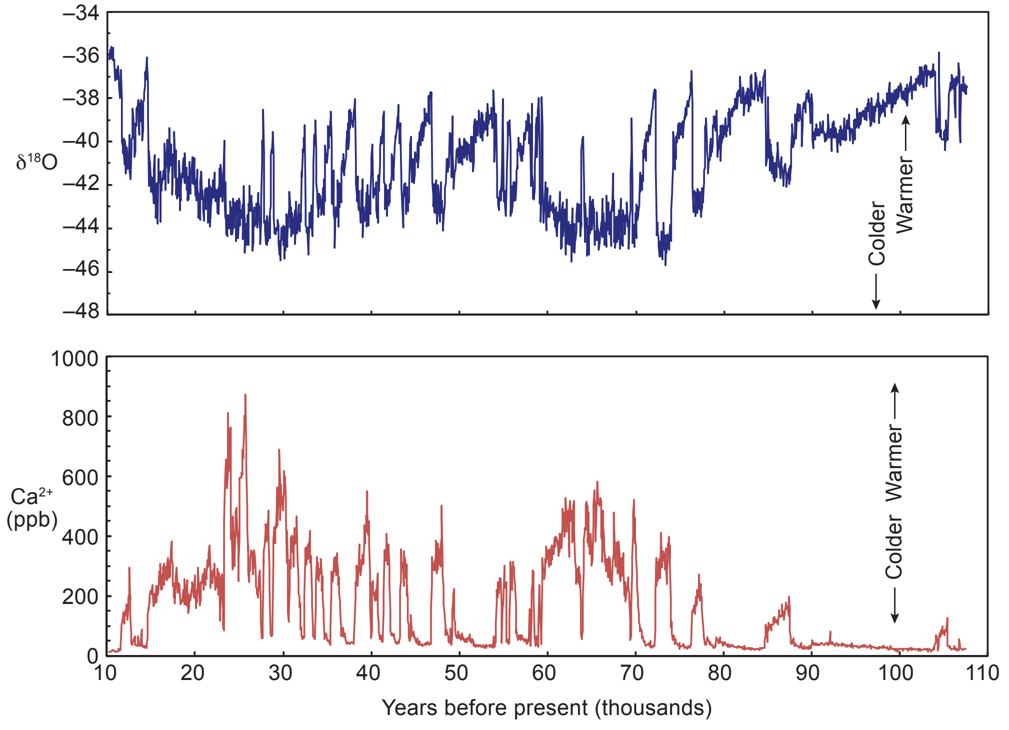

FIGURE 7.14 The records of $\delta^{18}O$ and Ca concentration in the NGRIP ice core

The ratio of oxygen-16 to oxygen-18 ($\delta^{18}O$) is a direct proxy for temperature, and the concentration of calcium (Ca) reflects the amount of dust in the atmosphere, which also varies with temperature because cold periods are drier and dustier than warm ones. Note the Dansgaard-Oeschger cycles, the abrupt swings between short, warm periods and longer, more stable cold periods. In all, there have been 25 D-O cycles. (Data from S. O. Rasmussen et al., "A Stratigraphic Framework for Abrupt Climate Changes During the Last Glacial Period Based on Three Synchronized Greenland Ice-Core Records: Refining and Extending the INTIMATE Event Stratigraphy," *Quaternary Science Reviews* 106 [2014]: 14–28; and I. K. Seierstad et al., "Consistently Dated Records from the Greenland GRIP, GISP2, and NGRIP Ice Cores for the Past 104 ka Reveal Regional Millennial-Scale $\delta^{18}O$ Gradients with Possible Heinrich Event Imprint," *Quaternary Science Reviews* 106 [2014]: 29–46)

proxy for temperature (figure 7.14).[34] The concentrations of calcium, sodium, and chlorine in the ice are also good, albeit indirect, climate indicators.[35] Calcium is present mainly in the form of carbonate and represents atmospheric dust. As the circulation that transports air to Greenland widens, which it does during cold periods, the winds scour dust from a wider landmass. Periods of cold climate are also drier, leading to more extensive arid regions and more dust. Sodium and chlorine come from sea salt. Both dust and sea salt reach Greenland primarily in the late winter to early spring because at that time of year, atmospheric circulation over North America is most intense and storms are more frequent than at other times.

Another important measurement in the ice is CH_4 concentration. Methane resides almost entirely in bubbles in the ice, and therefore the ice cores record the amount of CH_4 in the atmosphere. The CH_4 and temperature records vary together, and because the CH_4 signal originates mainly from wetlands in tropical regions, it has proved useful in chronological alignments of Antarctic and Greenland ice cores.

It is worth mentioning some other phenomena recorded in Greenland ice. Ice layers of high electrical conductivity indicate volcanic eruptions.[36] Electrical conductivity is mainly a measure of acidity, the most important ice component of which is sulfuric acid. Explosive volcanic eruptions can inject large amounts of sulfur dioxide into the atmosphere (chapter 6). Most of the major eruptions of the past 2,000 years show up in the ice, from the small eruption of Surtsey in Iceland (1963) to the gigantic eruption of Mount Tambora in Indonesia (1815). More ancient eruptions are also in evidence, such as that of Mount Mazama, which occurred around 5677 B.C.E. and created Crater Lake in Oregon.[37]

The ice cores also record human activities. In addition to indicating volcanic eruptions, the sulfate and nitrate contents of the ice document atmospheric pollution transported from the middle latitudes.[38] Their concentrations start to increase in ice formed in the late nineteenth century as the Industrial Revolution began to pick up steam. The passage of the Clean Air Act (1970) in the United States can also be detected as a reduction in atmospheric contaminants. Somewhat more arcane is the copper content of the ice, which documents pollution from copper smelting that began 2,500 years ago, about the time that coins began to be used in Greece, and continued into medieval times.[39]

MILLENNIAL-SCALE COLD–WARM CYCLES

Variations in temperature, amounts of dust and sea salt, and air composition reveal what is certainly the most remarkable feature of the Greenland ice cores: the sudden, dramatic swings between short, warm periods that lasted for hundreds to thousands of years and longer, and more stable cold periods during which temperatures on the Greenland ice cap were as much as 20°C (36°F) colder than they are today. The cycles are well recorded in the dust (calcium) record (see figure 7.14). Collectively, an astonishing 25 cycles, known as Dansgaard-Oeschger (D-O) cycles after the investigators who first recognized them, exist in the span of nearly 100,000 years.

The typical D-O cycle began with rapid warming in just a century or less during which air temperature rose 7 to 16°C (13–29°F).[40] This was followed by a gradual cooling that lasted for centuries to a couple of millennia, and then a very rapid cooling back to a period of cold that also continued for centuries to a few millennia. The pattern characterizes nearly all the cycles, even though their durations differed. The total magnitude of the temperature difference between cold and warm periods is about half that of the 100,000-year Pleistocene glacial–interglacial cycles. Although best defined in Greenland, the D-O cycles are also evident in other climate records for the Northern Hemisphere.[41]

What could have caused the D-O cycles? Certainly their durations are far too short to have been the result of orbital cycles. One idea is that the cause was changes in the North Atlantic meridional overturning circulation (MOC, the northward flow of warm surface water in the North Atlantic and the southward return of cold North Atlantic Deep Water [NADW] [chapter 2]). In this model, the cooling was brought about by the influx of freshwater into the North Atlantic due to the collapse of the ice sheets in the Northern Hemisphere.[42] Being less dense than saltwater, the freshwater spread out and covered the North Atlantic Ocean. This limited the loss of winter ocean heat and thus atmospheric warming, while flooding the ocean surface with more buoyant water that slowed or even stopped the production of NADW and therefore the MOC. This brings us to Antarctica.

THE LINK BETWEEN GREENLAND AND ANTARCTICA

Relatively large and abrupt shifts have also characterized the climate of Antarctica for the past several tens of thousands of years. Each warming in Antarctica started as Greenland descended into the very coldest part of its D-O period and well before Greenland's temperature abruptly increased, and the level of warming in Antarctica was generally proportional to the duration of the cold periods in Greenland.[43] These comparisons beg the question: How are the climates of Greenland and Antarctica linked?

One hypothesis that seeks to answer this question is known as the *bipolar seesaw*.[44] As explained, a prevailing theory to account for D-O events is that the influx of freshwater into the North Atlantic Ocean slowed or stopped the MOC and the production of NADW. The bipolar-seesaw model holds that the subsequent reduction in the amount of NADW reaching the South Atlantic resulted in the buildup of heat there and the northward migration of the Intertropical Convergence Zone. To appreciate this connection, one must keep in mind that the MOC is responsible for one-quarter of the global meridional heat transport. Furthermore, one might expect that the slower the overturning circulation, the longer the Greenland cold period, the more heat that built up in the South Atlantic, and the warmer it got in Antarctica, as observed. Slowly this hemispheric imbalance faded, eventually leading to the sudden reestablishment of overturning circulation, warming of the North Atlantic, and further cooling in the Southern Hemisphere. Thus the bipolar seesaw reflects both the processes in the North Atlantic and the inertial response of the South Atlantic to changes in the patterns of ocean and atmospheric circulation.[45]

THE TERMINATION OF THE LAST GLACIATION

The *Last Glacial Maximum* (LGM) is defined by a sea-level minimum (reflecting a maximum in ice volume) and occurred between about 30,000 and 20,000 years ago. The spread of ice leading up to the LGM was driven primarily by a decrease in insolation in the high northern latitudes (figure 7.15). The glacial termination began abruptly between about 19,000 and 18,000 years ago. It was initiated by a rise in Northern Hemisphere insolation that had started 2,000 or 3,000 years earlier. As can be clearly seen in figure 7.15, the increase in insolation was followed, 1,000 or so years later, by an abrupt and rapid growth in atmospheric CO_2 content, and this is what must have caused the dramatic warming that followed.[46] As noted, on the millennial scale, the CO_2 content of the atmosphere appears to be tied to the Southern Ocean, where overturning circulation currently transfers huge amounts of CO_2 from the atmosphere to the deep ocean.[47] Exactly what caused the overturning circulation patterns of the Southern Ocean to change at the end of the last glaciation and bring about the increase in CO_2 uptake remains unclear.

The warming that began in earnest about 18,000 years ago led to prolonged glacial retreat, the eventual disappearance of the ice sheets in North America (termed the Laurentide) and

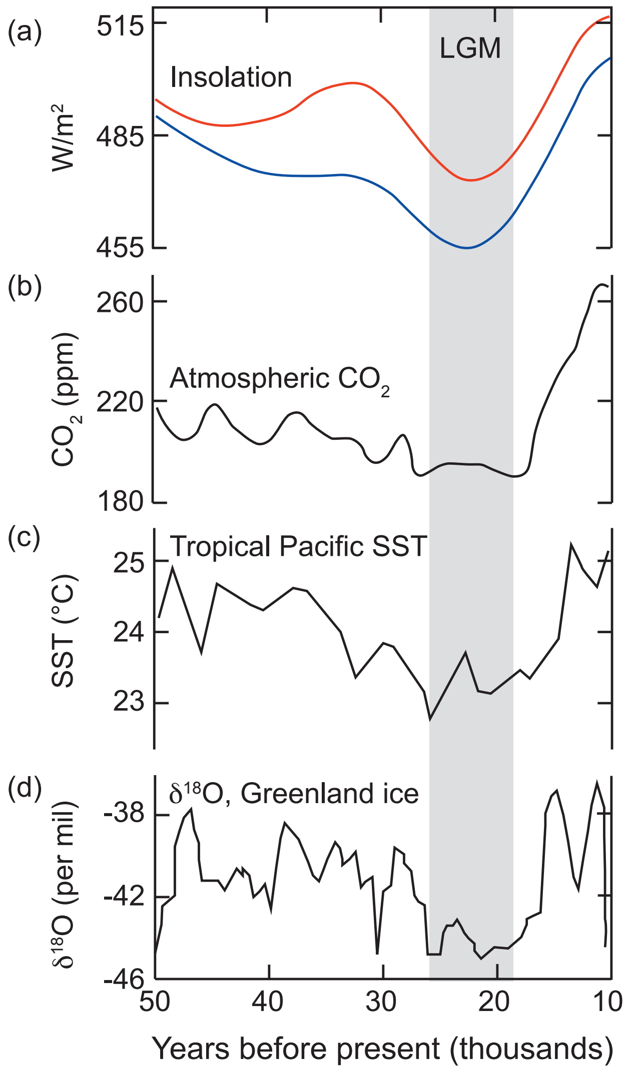

FIGURE 7.15 Changes in conditions associated with the close of the last glaciation
(*a*) Northern Hemisphere insolation at 65°N (*blue*) and 45°N (*red*) latitudes; (*b*) atmospheric CO_2 content as recorded in Antarctic ice cores; (*c*) sea-surface temperature of the tropical Pacific; and (*d*) the oxygen-isotope composition of Greenland ice. Note that CO_2 rise and rapid warming lagged behind the increase in Northern Hemisphere insolation. The vertical gray bar indicates the Last Glacial Maximum. (Redrawn from P. U. Clark et al., "The Last Glacial Maximum," *Science* 325 [2009]: 710–714, where references to the original sources may be found)

Scandinavia (figure 7.16), and the dwindling of those on Greenland and Antarctica. The loss of ice caused a rapid rise in sea level that stabilized only 12,000 years later (chapter 8).

THE YOUNGER DRYAS AND THE BEGINNING OF THE HOLOCENE

The *Younger Dryas* (YD) is a 1,200-year interval of cold that occurred between 12,900 and 11,700 years ago (figure 7.17; see figure 7.14),[48] after which the current geologic epoch, the

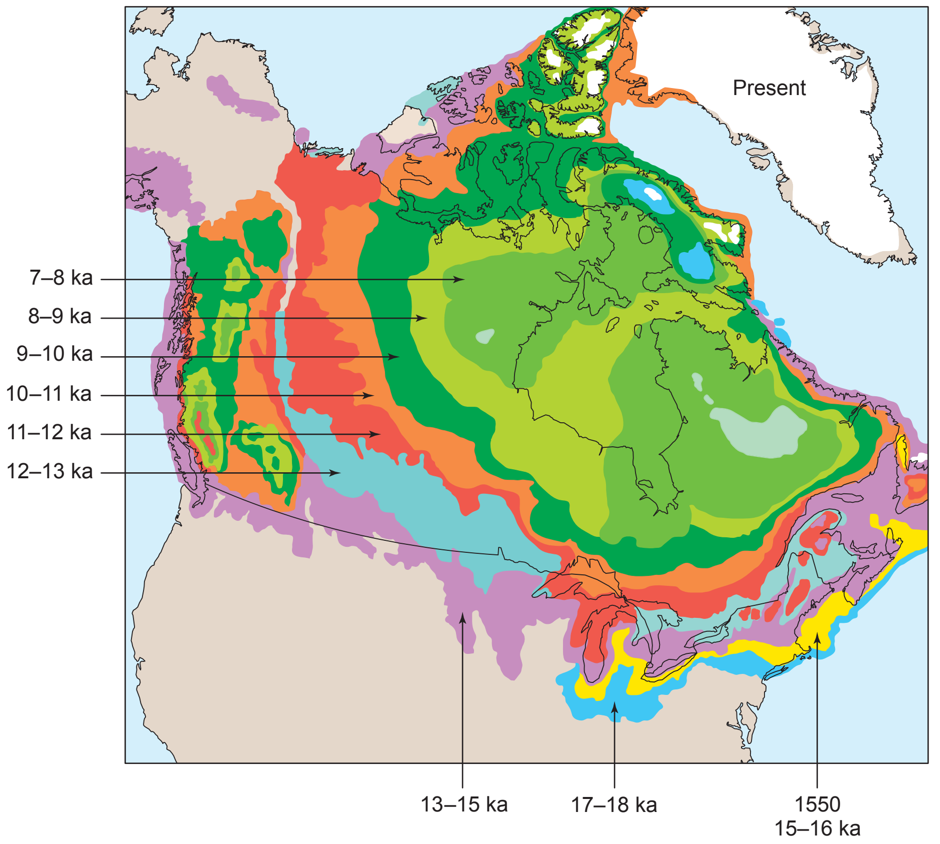

FIGURE 7.16 Retreat of the Laurentide Ice Sheet
The Laurentide Ice Sheet, which covered most of Canada and the northern United States, retreated slowly over 18,000 years. (Redrawn from S. Dutch, "Pleistocene Glaciers and Geography," Department of Natural and Applied Sciences, University of Wisconsin–Green Bay, https://www.uwgb.edu/dutchs/EarthSC202Notes/GLACgeog.HTM)

Holocene, began. In Greenland, the transition from warm to cold conditions at the beginning of the YD took about a century. In contrast, the end of the YD was marked by an increase in temperature of 10 ± 4°C (18 ± 7°F) and a doubling of the rate of snow accumulation in less than a decade, possibly in just a few years. The abruptness and magnitude of this change are far greater than any that have occurred since.

What may have caused the YD?[49] One hypothesis is that the Laurentide ice sheet began a rapid retreat as warming proceeded before the YD. The retreat of the ice sheet introduced meltwater into a quickly growing Lake Agassiz, an enormous body of water covering most of Manitoba and western Ontario. Lake Agassiz originally drained down the Mississippi River into the Gulf of Mexico, but the warm conditions began to open the ice dams blocking other drainages. In one catastrophic breach, the water level of Lake Agassiz fell some 40 meters (130 feet) in just a few years by spilling a vast amount of freshwater directly into the North Atlantic Ocean, possibly through Hudson Bay. This shut down the MOC and initiated glacial conditions, but when the amount of freshwater flowing into the North Atlantic

FIGURE 7.17 ***The flower of* Dryas octopetala, an Arctic-alpine evergreen shrub**
Dryas octopetala, because of its abundance in Younger Dryas sediment, gave its name to the Dryas periods. (Griensteidl, https://commons.wikimedia.org/wiki/File:Dryas_octopetala_Tscheppaschlucht2.jpg)

dwindled, warm ocean water was eventually able to reclaim the ocean surface. A sudden, northward shift of the generally westerly winds due to the melting of sea ice and the warming of ocean-surface waters may have brought warm and moist air to Greenland and the abrupt end of the YD.[50]

An interesting event occurred several thousand years later, supporting the idea that an influx of freshwater into the North Atlantic can upset the MOC and cause cooling. The Greenland ice cores show a period of unusually cold climate beginning about 8,500 years ago. It lasted for several hundred years and represents the coldest period in the entire Holocene. Fossil and other evidence from North Atlantic sediment cores indicate that two massive outpourings of freshwater flooded the North Atlantic, one 8,490 years ago and the other 8,290 years ago.[51] The timing is coincident with the final drainage of Lakes Agassiz and Ojibway, the two lakes in central Canada that remained from the earlier glacial floods.

A Lesson from the Distant Past: The Paleocene–Eocene Thermal Maximum

We have stepped through many of the major events in Earth's climatic history over the past 5 million years that bring us to the beginning of the Holocene, the most recent epoch of geologic history (chapter 8). As we have seen, events since the Pliocene are replete with fascinating and abrupt climate changes that have been driven by many processes within the climate system as well as by external forcings. As interesting and varied as that history is, however, nowhere does it contain an event remotely analogous to the changes that are occurring today and are expected to continue throughout the twenty-first century. For that, we must go back 56 million years to one of Earth's most rapid and abrupt natural climate changes.

During the late Paleocene, climate had been slowly warming, but then an enormous mass of carbon suddenly flooded the ocean and atmosphere.[52] In perhaps 25,000 years, approximately 10,000 gigatons (Gt [1 Gt = 1 billion metric tons]) of carbon entered the climate system,[53] an amount similar to that which human activities would add to the climate system over the next millennium at the present rate of emissions. Along with increasing atmospheric CO_2 content, the average global surface temperature rose 4 to 5°C (7–9°F),[54] and the ocean acidified.[55] The event is known as the Paleocene–Eocene Thermal Maximum (PETM), and it lasted for about 200,000 years.

The changes during the PETM had a dramatic effect on terrestrial and marine ecosystems. For example, we know from the fossil record that mammalian diversification surged and body sizes evolved to become smaller, and the ranges of mid-latitude flora moved poleward by many hundreds of kilometers.[56] In the ocean, 35 to 50 percent of the benthic foraminifera species disappeared, but planktonic foraminifera diversified and shifted their ranges to higher latitudes.[57]

The geologic evidence for the PETM is remarkably displayed by deep-sea sediment cores.[58] In them is found carbonate-rich mud ("carbonate ooze") from the Paleocene overlain at a sharp boundary by red clay from the Eocene, and then the red clay gradually grades back to carbonate ooze upward through the younger layers. The boundary between the carbonate ooze and the red clay is accompanied by a sudden decrease in $\delta^{13}C$ (a measure of the ratio of the isotope carbon-13 to carbon-12). The decrease is indicative of the addition to the ocean of a large quantity of carbon with a somewhat lower $\delta^{13}C$ value than that of marine carbonates (figure 7.18).

The transition from carbonate ooze to red clay indicates that the ocean acidified (chapter 4); that is, ocean pH and carbonate-ion concentration (CO_3^{2-}) decreased. The change in ocean composition caused a rise in the carbonate compensation depth (CCD) (chapter 4). Recall that above this depth, calcareous sediments such as chalk and limestone form as the shells of planktonic foraminifera accumulate on the ocean floor, while below the CCD, carbonate shells dissolve before they can be buried and the sediment consists of only red clay and the shells of silica plankton (for example, diatoms and radiolarians). The marine-sediment records suggest that during the PETM, the CCD shoaled by more than 2,000 meters (6,600 feet).[59]

Where did all the carbon come from? There are plenty of ideas. One hypothesis is that the carbon originated from methane hydrate (chapter 9) that broke down as the ocean warmed.[60] Consistent with this idea is that the PETM was preceded by a period of gradual global warming, which would account for the ocean warming.[61] Another proposal is that the carbon was dominantly of volcanic origin.[62] As it turns out, the age of the PETM corresponds precisely with that of an enormous outpouring of lava constituting what is known as the North Atlantic Igneous Province. The outpourings were associated with the breakup of

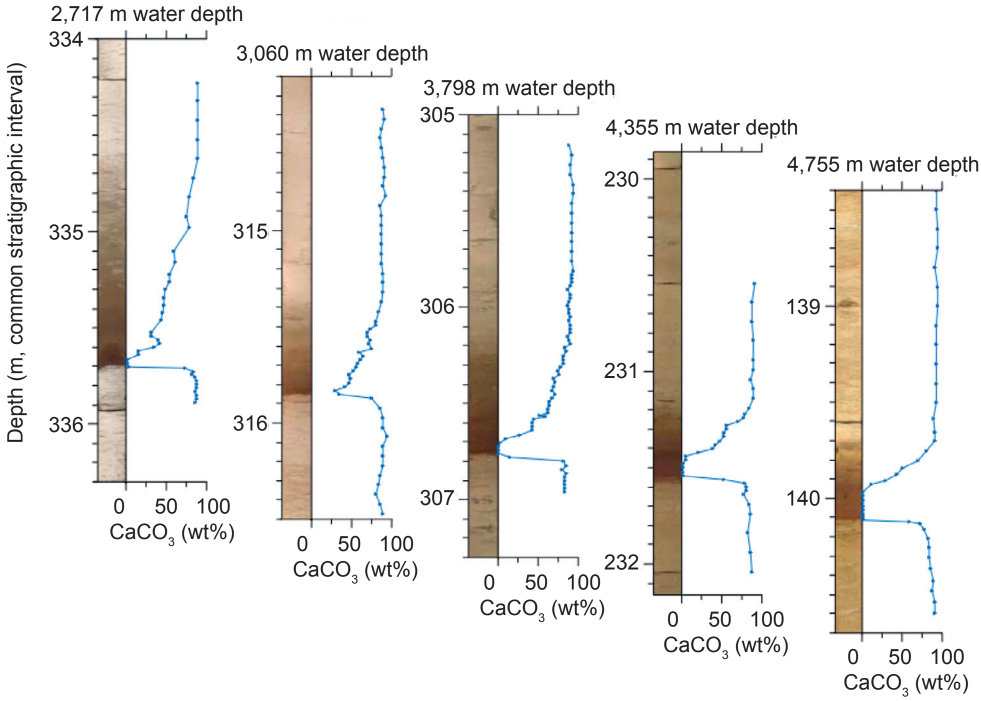

FIGURE 7.18 The record of the Paleocene–Eocene Thermal Maximum in deep-sea sediment cores
The sediment cores display an abrupt change in which Paleocene carbonate-rich muds are overlain by Eocene darker red clays, and then the red clays gradually grade back to carbonate mud upward through the stratigraphic section. The boundary is accompanied by a sudden decrease in the ratio of carbon-13 to carbon-12 (not shown), indicative of the addition of organic carbon to the ocean–atmosphere system. The sudden increase in atmospheric CO_2 caused the oceans to become more acidic and the carbonate compensation depth to rise to shallower levels. The "body stage," during which conditions remained warm and atmospheric CO_2 content remained high, and the "recovery stage," during which conditions slowly returned to normal, are shown. Water depths are for the tops of the sediment cores. (Redrawn from J. C. Zachos et al., "Acidification of the Ocean During the Paleocene–Eocene Thermal Maximum," *Science* 308 [2005]: 1611–1615)

the North Atlantic Ocean and are now mainly represented by basaltic lavas and sills found along and off the coasts of southeastern Greenland and western Scotland. Gases associated with the erupting lavas likely contained CO_2 as a major component.

Throughout much of the PETM, the climate remained warm and the level of atmospheric CO_2 remained high. The return to cooler conditions occurred over a period of about 80,000 years.[63] During this recovery time, carbon presumably was removed by weathering of rocks (chapter 4) and burial in marine sediments.[64] The time it took the climate system to recover during the PETM could be a measure of the time in which the carbon cycle can return conditions to normal. Although human activities are not going to introduce the 10,000 Gt of carbon emissions associated with the PETM anytime soon, the *rate* at which carbon is presently being introduced into the atmosphere is several times greater than it was during the onset of the PETM (Back-of-the-Envelope Calculation).[65] Even if the magnitudes of the emissions are not the same, are the changes that took place during the PETM over thousands of years a harbinger for climate and ecosystem changes that may take place over the next millennium? Some think they are.[66]

BACK-OF-THE-ENVELOPE CALCULATION
Carbon Emissions Today and at the PETM

The PETM was marked by a massive and abrupt release of carbon into the atmosphere accompanied by a rapid warming of the climate. But how rapid was the carbon release compared with the current rate caused by human activities?

It appears that the PETM was brought about by the release of about 10,000 gigatons of carbon (Gt C) into the atmosphere and ocean. How much is that? A gigaton (Gt) is 10^{15} grams, or 1 billion metric tons, which is hard to imagine. To put it in perspective, consider that a Nimitz-class aircraft carrier, the largest warship ever built, weighs a measly 88,000 metric tons under full load.[1] A single gigaton would therefore amount to about 11,364 of these warships! If you prefer an example from land, the Great Pyramid of Giza is estimated to weigh about 5.9 million metric tons,[2] requiring 170 such pyramids to equal just 1 gigaton. In either case, these quantities put into perspective the massiveness of a single gigaton and therefore the huge amount of carbon that was emitted into the atmosphere during the PETM (some 1.7 million pyramids worth!).

Now let's consider the *rate* of the carbon release during the PETM. The most recent estimates suggest that it happened over about 25,000 years. Using the total carbon-release and time-span estimates, and assuming that the emission rate was constant, the rate at which carbon was released into the atmosphere during the PETM was 0.4 Gt C a year. How does that compare with what is happening now? The Intergovernmental Panel on Climate Change estimates that human activities have resulted in 555 ± 85 Gt C released into the atmosphere between 1750 and 2011. While that is not nearly the total amount of carbon released during the PETM, it indicates that the average rate of carbon emission since the beginning of the Industrial Revolution is between 1.79 and 2.44 Gt C a year. But the time period that we have considered includes the very beginning of the Industrial Revolution, when emissions were much lower than they are today. In fact, global emissions over the past decade have been between 8 and 9 Gt C a year, or 20 to 22.5 times greater than the rate that we have estimated for the PETM. In other words, we have not yet reached the total carbon release of the PETM, but if we continue at current rates, we would reach the magnitude of the PETM release much faster. Even if we assume an average rate equal to the smaller values estimated over the full 1750–2011 period, it would take us only between 4,098 and 5,586 years from the beginning of the Industrial Revolution to reach the total emissions of the PETM. If we assume the higher rates of 8 to 9 Gt C a year, it would take just 1,111 to 1,250 years.

The PETM is considered to be one of the closest past analogues for what Earth is experiencing today. On geologic timescales, the rate of carbon release into the atmosphere happened blazingly fast and yielded massive changes across the globe. And yet, if we compare the rates of carbon release during the PETM with those of today, we find ourselves vastly outpacing even the PETM. While we have a long way to go if we are ever to reach the magnitude of carbon released during that period, it should serve as a cautionary reminder that we are releasing massive amounts of carbon into the atmosphere much faster than Earth ever did without our help during the PETM.

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1. United States Navy, "Aircraft Carriers," January 31, 2017, http://www.navy.mil/navydata/fact_display.asp?cid=4200&tid=200&ct=4.
2. "Great Pyramid of Giza," Wikipedia, https://en.wikipedia.org/wiki/Great_Pyramid_of_Giza.

KEY POINTS IN THIS CHAPTER

This chapter presents numerous examples of the way paleoclimate studies have given us insight into how the climate system works. The discussion should also make clear that climate operates on a variety of temporal scales, some of which are much longer than the human lifetime. This means that we must understand climate change not through the lens of human experience but through the more abstract one of how the planet "works."

Over geologic time, climate has varied dramatically, from periods of deep cold to times when both global temperature and the CO_2 content of the atmosphere were higher than they are today. These changes reflect natural changes in the larger Earth system, in contrast to the current anthropogenic warming, and indicate that the world was a very different place during those periods of increased temperatures and elevated levels of CO_2. The paleoclimate record also attests to the fact that climate has remained remarkably stable for roughly the past 11,000 years, since the end of the Younger Dryas. Complex, advanced civilizations emerged only during this time. While Earth will surely survive dramatic future shifts in climate, the extent to which human civilization will be disrupted by, or be able to adapt to, the current anthropogenic climate change remains an open question.

To summarize what we have learned:

1. Paleoclimatology gives insight into how climate works across differing timescales and thus how it works today and may change in the future.

2. Earth's latest ice age began about 2.6 million years ago. The concept of an ice age developed in the early nineteenth century from observations of glacial erratics and other features in the valleys of the Swiss Alps. That great ice sheets covered large parts of Europe and North America is indicated by the extensive deposits of till left by the ice.

3. Earth's obliquity, eccentricity, and precession oscillate in cycles of approximately 41,000, 100,000, and 26,000 years, respectively. According to Milankovitch theory, the interplay of the three cycles creates variations in insolation during the summer in the high latitudes of the Northern Hemisphere, which, in turn, accounts for the waxing and waning of the ice sheets during the Pleistocene.

4. Stable isotopes of oxygen, hydrogen, and carbon (and, less so, nitrogen and sulfur) are important tools with which climate records are extracted, in combination with chronological data, from sediment cores, ice cores, and other geologic records.

5. The climate of the Pliocene, from about 5 to 3 million years ago, was relatively warm and stable compared with the climate since. The climate of that time may be an analogue to what climate could become by the end of the twenty-first century. One question is why the climate of the Pliocene was warmer than today's, even though the external forcings were not so different. A second is why, beginning about 3.1 million years ago, the Pliocene climate began to cool.

6. The Pleistocene, which began about 2.6 million years ago, marks the emergence of well-defined, 41,000-year glacial–interglacial cycles, reflecting the amplification of Earth's obliquity cycle. They continued, along with a steady cooling, until about 1.2 million years ago,

when the mid-Pleistocene Transition brought us into the world of 100,000-year glacial–interglacial cycles.

7. The glacial–interglacial cycles of the past 800,000 years are characterized by orbitally driven glacial periods that typically last for 80,000 to 90,000 years and short interglacials of about 10,000 years' duration. The cold periods ended abruptly and the warm periods more gradually, in a sawtooth pattern of temperature fluctuations.

8. For the past 800,000 years, atmospheric CO_2 has varied between about 170 and 300 parts per million (ppm) in nearly precise step with the glacial–interglacial cycles. This implies a natural control on atmospheric CO_2 content, probably involving the exchange of CO_2 between the ocean and the atmosphere in the Southern Ocean.

9. At more than 400 ppm, the CO_2 content of the present-day atmosphere is higher than it has been in at least 800,000 and possibly in more than 3 million years.

10. Greenland ice cores hold a detailed record of North Atlantic climate extending back through the Eemian, more than 136,000 years ago.

11. The last glacial period contains numerous Dansgaard-Oeschger (D-O) cycles and demonstrates the central role of the North Atlantic meridional overturning circulation (MOC) in controlling climate around the North Atlantic. The fluctuations in the MOC also link the climate of Antarctica to that of the North Atlantic through the bipolar seesaw.

12. The Last Glacial Maximum (LGM) ended abruptly about 18,000 to 19,000 years ago due to an increase in insolation during the summer in the Northern Hemisphere, amplified by a rise in the CO_2 content of the atmosphere. This led to the prolonged retreat and eventual disappearance of the Laurentide and Scandinavian ice sheets.

13. The 8,000-year transformation from the LGM to the Holocene proceeded in fits and starts. It included the Younger Dryas (YD) cold period, which resulted from changes in the dynamics of the climate system rather than external forcings. The end of the YD marked the beginning of the Holocene, the current geologic epoch.

14. The Paleocene–Eocene Thermal Maximum (PETM) was a 200,000-year period of significant warmth and ecosystem disruption that began abruptly about 56 million years ago and ended only gradually over many thousands of years. It could be an analogue for our current and future climate perturbations.